Pattern Generator

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Objectives

- Giving info on various types of patterns and the combinations which are possible to make with the help of loops.
- To generate 12 distinct pattern variations across 3 types and 4 shapes.
- Using Procedural Oriented Programming approach and using nested loops and iterative statements to generate a wide variety of patterns.
- To produce a readable, well structured and well commented c++ source code.

User input

```
#include <iostream>
using namespace std;

int main()
{
    int i,j,t,c;
    cout<<"----------------------------------------------"<<endl;
    cout<<"Enter the type of Pattern you want"<<endl;
    cout<<"----------------------------------------------"<<endl;
    cout<<" 1. Star Pattern \n 2. Number Pattern \n 3. Continuous Number Pattern"<<endl;
    cout<<"Enter your choice: ";
    cin>>t;
    if(t==1)
    {
        cout<<"----------------------------------------------"<<endl;
        cout<<"Enter the shape of Pattern you want"<<endl;
        cout<<"----------------------------------------------"<<endl;
        cout<<" 1. Square \n 2. Triangle \n 3. Left Triangle \n 4. Right Triangle"<<endl;
        cout<<"Enter your choice: ";
        cin>>c;
        cout<<"----------------------------------------------"<<endl;
```

Star Patterns

```
if(c==1)
{
    for(i=1;i<=5;i++)
    {
        for(j=1;j<=5;j++)
        {
            cout<<"* ";
        }

        cout<<endl;
    }
}
if(c==2)
{
    for(i=5;i>=1;i--)
    {
        for(j=1;j<=5;j++)
        {
            if(j>=i)
            cout<<"* ";
            else
            cout<<" ";
        }
        cout<<endl;
    }
}
```

```
if(c==3)
{
    for (i=1;i<=5;i++)
    {
        for (j=1; j<=i;j++)
        {
            cout << "* ";
        }
        cout<<endl;
    }
}
if(c==4)
{
    for (i=1;i<=5;i++)
    {
        for (int k=5-i;k>0;k--)
            cout << " ";
        for(j=1;j<=i;j++)
        {
            cout<<"*";
        }
        cout<<endl;
    }
}
}
```

Number Patterns

```
if(t==2)
{
    cout<<"-----------------------------------------------"<<endl;
    cout<<"Enter the shape of Pattern you want"<<endl;
    cout<<"-----------------------------------------------"<<endl;
    cout<<" 1. Square \n 2. Triangle \n 3. Left Triangle \n 4. Right Triangle "<<endl;
    cout<<"Enter your choice: ";
    cin>>c;
    cout<<"-----------------------------------------------"<<endl;
    if(c==1)
    {
        for(i=1;i<=5;i++)
        {
            for(j=1;j<=5;j++)
            {
                cout<<i<<" ";
            }
            cout<<endl;
        }
    }
    if(c==2)
    {
        for (i=1;i<=5;i++)
        {
            for (j=1;j<=5-i;j++)
            {
            cout << " ";
            }
            for (j=1;j<=i;j++)
            {
            cout << i << " ";
            }
        cout << endl;
        }
```

```
if(c==3)
{
    for (i=1;i<=5;i++)
    {
        for (j=1; j<=i;j++)
        {
            cout << i;
        }
        cout<<endl;
    }
}
if(c==4)
{
    for (i=1;i<=5;i++)
    {
        for (int k=5-i;k>0;k--)
            cout << " ";
        for(j=1;j<=i;j++)
        {
            cout<<i;
        }
        cout<<endl;
    }
}
```

Continuous Number Patterns

```cpp
if(t==3)
{
    cout<<"----------------------------------------------"<<endl;
    cout<<"Enter the shape of Pattern you want"<<endl;
    cout<<"----------------------------------------------"<<endl;
    cout<<" 1. Square \n 2. Triangle \n 3. Left Triangle \n 4. Right Triangle"<<endl;
    cout<<"Enter your choice: ";
    cin>>c;
    cout<<"----------------------------------------------"<<endl;
    if(c==1)
    {
        int num = 1;
        for(i=1;i<=3;i++)
        {
            for(j=1;j<=3;j++)
            {
                cout<<num<<" ";
                num++;
            }
            cout<<endl;
        }
    }
    if(c==2)
    {
        int num=1;
        for (i=1;i<=3;++i)
        {
            for (int k= 1; k<=3 - i; ++k)
            {
                cout <<" ";
            }
            for (int j = 1; j <= (2 * i - 1); ++j)
            {
                cout<< num++;
            }
            cout << endl;
        }
    }
```

```cpp
if(c==3)
{
    int num=1;
    for (i=1;i<=5;i++)
    {
        for(j=1;j<=i;j++)
        {
            cout<<num<< " ";
            num++;
        }
        cout << endl;
    }
}

if(c==4)
{
    int num=1;
    for (i=1;i<=4;i++)
    {
        for (int k=4-i;k>0;k--)
            cout << "  ";
        for(j=1;j<=i;j++)
        {
            cout<<num<<" ";
            num++;
        }
        cout<<endl;
    }
}
```

Conclusion

The Pattern Generator Program is a successful implementation of a menu driven C++ console application that demonstrates the practical use of loops, conditional statements, sequential counters, and basic console I/0.

Through this project, the core programming concepts of loop nesting, variable scoping, and flow control were applied in a visual and intuitive way.

This project strengthened output positioning to produce two-dimensional patterns on a one-dimensional output stream.
The skills and logic patterns applied here form a strong foundation for more complex programming tasks such as matrix operations, game grids, or GUI layout calculations.